1. Authentication & Profile

- **Doctor Login:** Secure login using email/phone, OTP, and fingerprint or face unlock.
- **Doctor Profile:** Manages educational degrees, specialty, BMDC registration number, chamber/hospital details, and a digital signature for prescriptions (no seal).
- **Hospital-Fenced Biometric Attendance:** Geofenced check-in and check-out module that verifies doctor attendance only when physically present inside hospital premises, validated via on-device biometrics (fingerprint or face authentication).

2. Patient Management & EHR

- **Patient Directory & Search:** Instant search by patient name, mobile number, or unique Patient ID.
- **Auto Patient Account Creation:** Creating a prescription for a new patient automatically registers and generates a primary patient account in the system.
- **Patient Details & History:** Unified profile showing complete medical history, previous consultations, and past prescription logs.
- **Adherence Insights:** Tracks whether the patient is taking medications on time via adherence percentages and interactive calendar heatmaps.
- **Private Clinical Notes:** Doctor-only scratchpad for internal clinical observations and follow-up flags hidden from the patient view.

3. Digital Prescription Builder

- **Smart Medicine Auto-Suggest:** Quick-search medication library auto-completing brand and generic names, forms (tablet, syrup, etc.), and strengths.
- **Dosing & Schedule Configurator:** Granular selectors for exact dosage, frequency (e.g., 1+0+1, before/after meals), duration, and specific instructions.
- **Voice-to-Text Instructions:** Dictation tool allowing physicians to speak instructions directly into prescription fields in English or Bangla.

- **Test & Diagnostic Attachments:** Recommends required pathological tests and allows review of previously uploaded test reports.

4. Prescription Generation, QR Sync & Access Control

- **Single Account Active Prescription Ownership:** Each prescription remains active strictly under one primary patient account at a time to prevent duplicate dosing conflicts.

- **Family / Caregiver Member Access (Read-Only Sharing):** Primary account holders can link family members or caregivers to their profile, granting them secure view-only access to active prescriptions and medication schedules.

- **Prescription Summary & QR Code:** Generates a dynamic QR code on the prescription draft so the patient can scan it to instantly claim and sync the prescription into their account.

- **PDF Prescription:** Creates clean, printable PDF prescriptions with doctor information and the digital signature.

5. Doctor Analytics & Reports

- **Prescription Statistics:** Dashboard tracking daily and monthly patient counts, new versus follow-up visits, and total prescriptions issued.

- **Patient Adherence Statistics:** Aggregated overview of patient compliance rates to track adherence to prescribed treatments.

- **PDF/CSV Export:** Exports clinical summaries, analytics, and prescription logs in CSV or PDF format.

6. App Preferences & System

- **Bilingual Support:** Full app and prescription template support for both English and Bangla.

- **Light/Dark Theme:** System-friendly Light and Dark display modes for comfortable viewing.

1. User & Account Management

- **User & Patient Management:** View, filter, and manage all registered platform users and patient profiles, including account details and contact information.
- **Doctor Management:** Directory to monitor registered physicians, verify BMDC numbers, specializations, degrees, and affiliated chambers.
- **Account Verification:** Review uploaded documentation and credentials to manually approve or reject new doctor and facility accounts.
- **Suspension & Deletion:** Administrative controls to temporarily suspend or permanently delete user and doctor accounts violating platform policies.

2. Clinical & Prescription Oversight

- **Prescription Monitoring:** Platform-wide oversight to audit issued digital prescriptions, track activity logs, and flag anomalies or misuse.

3. Content & AI Knowledge Base

- **Announcement Management:** Create, publish, and schedule system-wide alerts, banner announcements, and targeted notices for users or doctors.
- **Chatbot Knowledge Base:** Portal to update medical reference datasets, configure symptom-response matrices, and fine-tune AI medical chatbot behaviors.

4. Push Notifications & Communication

- **User-Group Notifications:** Broadcast custom alerts, public health notices, and system announcements segmented by user role (all users, patients only, or doctors only).

5. System Analytics & Reports

- **User Growth Analytics:** Visual metrics tracking daily, weekly, and monthly acquisition trends for patients and medical practitioners.
- **Prescription Analytics:** Performance graphs showing total prescriptions written, QR code redemption volume, and regional distribution.

- **Adherence Analytics:** High-level overview tracking overall patient medication compliance rates and regimen completion across the platform.
- **PDF & CSV Reports:** One-click data export tool generating analytical, operational, and audit logs in PDF and CSV formats.

6. Security & System Administration

- **System Logs:** Centralized audit logs capturing application errors, server activity, admin actions, and security-relevant events.
- **Roles & Permissions:** Role-based access control (RBAC) to define granular permissions for sub-admins, moderators, and support agents.
- **Maintenance Mode:** Global toggle switch to place the platform into maintenance mode with a user-facing notice during upgrades or downtime.

Developer Instruction

1. Medication Management

- **Medication Dashboard:** Displays current medicines, dosages, timings, and live status buttons (Taken, Missed, Upcoming).

- **Prescription Entry (3 Ways):**

    - **QR Code Scan:** Instantly imports all medicine details by scanning the QR code on a registered doctor's prescription.
    - **OCR Camera Scan:** Scans and reads printed or handwritten text from non-registered prescriptions using the camera.
    - **Manual Entry:** Lets users type in medicine names, dosages, and schedules by hand.

- **Adherence Tracking:** Visual calendar and heatmap views to track daily medicine intake consistency and streaks.

2. Scheduling & Smart Alerts

- **Intelligent Notifications:** Timely medicine reminders that allow users to mark doses as Taken or Snooze directly from the lock screen.

- **Inventory & Refill Alerts:** Tracks remaining pill counts, alerts when supplies run low, and reminds users to refill from the pharmacy.

- **Immunization Tracker:** Complete vaccination schedules for kids and adults with automatic alerts for upcoming shots.

3. Clinical & Diagnostic Tools

- **AI Medical Chatbot:** Conversational health assistant with voice-to-text input to answer health questions and check symptoms.

- **Pediatric Dose Calculator:** Calculates safe, accurate medicine doses based on a child's exact age and weight.

- **Medical Center Locator:** GPS tool to locate nearby hospitals, clinics, and pharmacies on a map with direct contact options.

4. Blood Bank & Donor Network

- **Blood Bank Directory:** A map and list of registered blood banks showing live blood group availability and contact numbers.

- **Donor & Recipient Hub:**

    - **Unified Profile:** No separate account required. Users simply turn on a "Blood Donor" toggle inside their existing profile to add blood group, last donation date, and availability.

    - **Emergency Broadcast & Direct Call:** In an emergency, requests trigger instant push alerts to matching nearby donors, with a button to call the donor directly from the app.

5. Reports, System & Security

- **Health Records & Export:** Generates clean PDF reports of prescription history and medicine logs to share with doctors.

- **Cloud Backup & Restore:** Encrypted cloud backup to keep health data safe and restore it easily across devices.

- **Language & Theme Support:** Full support for both English and Bangla, along with Light and Dark display modes.

- **Biometric Authentication:** Protects sensitive medical records using fingerprint scan and face unlock.

1. Pharmacy Registration & Verification

- **Pharmacy Registration & Profile:** Onboarding portal to register pharmacy details including pharmacy name, address, contact information, and map/GPS location.
- **License & Pharmacist Credentials:** Input portal for drug license numbers, official registration documents, and assigned pharmacist information.
- **Admin Verification:** Verification pipeline where the platform super-admin reviews uploaded licenses and credentials to approve and activate the pharmacy account.

2. Pharmacy Operations Dashboard

- **Today's & Pending Prescriptions:** Real-time counters showing total incoming prescriptions received today and active prescriptions waiting to be fulfilled.
- **Dispensing & Inventory Alerts:** Live tracking of medicines dispensed, alongside instant alert counters for low-stock drugs and items nearing expiration.

3. Prescription Management (Core Module)

- **Prescription Inbox:** Centralized incoming feed of digital prescriptions sent by patients or routed through the platform.
- **Prescription ID Search & QR Scan:** Fast retrieval tool allowing pharmacists to look up prescriptions via unique Prescription ID or scan patient prescription QR codes directly with a camera.
- **View Prescribed Medicines:** Detailed display of all prescribed medicines, generic compositions, dosages, strengths, frequencies, and duration instructions.
- **Prescription Status Workflow:** Step-by-step lifecycle management for each order:
    - **Pending:** Newly received, awaiting staff review.
    - **Processing:** Medicines being picked, verified, and packaged.
    - **Dispensed:** Medication handed over or delivered to the patient.
    - **Cancelled:** Order rejected or cancelled due to lack of stock or customer request.

4. Medicine Master Data

- **Medicine Specifications:** Catalog capturing brand name, generic name, strength, and dosage form (e.g., tablet, capsule, syrup, injection).

- **Manufacturer & Pricing:** Records manufacturer details, unit retail price (MRP), and applicable discount rules.
- **Stock & Batch Details:** Tracks available stock quantities, unique batch numbers, and assigned expiration dates.

5. Inventory & Stock Control

- **Add & Update Stock:** Data entry forms to log incoming supplier stock shipments and manually adjust stock quantities.
- **Stock History & Batch Tracking:** Audit trail recording inventory additions, dispensing deductions, batch movements, and return logs.
- **Low-Stock & Expiry Alerts:** Automated notifications triggered when stock levels hit critical thresholds or batches approach expiration within a set time frame.
- **Supplier Management:** Contact registry storing medicine distributor and supplier company profiles, agent contacts, and invoice reference numbers.

1. Hospital Registration & Verification

- **Hospital Registration & Profile:** Onboarding portal to register the facility with name, hospital type, contact details, and map location.

- **License & Admin Verification:** Document upload portal for official medical licenses and health authority approvals, reviewed and activated by the platform super-admin.

2. Hospital Operations Dashboard

- **Real-Time Operational Metrics:** Live counter showing total registered patients, today's scheduled appointments, and on-duty doctors currently available.

- **Activity Summary:** Real-time stream of completed consultations, new appointment bookings, and general daily operational workflows.

3. Department Management

- **Department Directory & Setup:** Tools to create, edit, and organize hospital departments (e.g., Cardiology, Pediatrics, Orthopedics).

- **Staff Mapping & Contacts:** View and assign doctors to specific departments alongside their respective extension lines and room allocations.

4. Doctor Management

- **Doctor Profiling:** Registry to add doctors with their BMDC registration number, qualifications, specialization, and assigned department.

- **Doctor Availability & Rostering:** Duty roster system to configure shift timings, consultation hours, weekly visiting days, and leave statuses.

5. Patient Management & Records

- **Patient Search & Profile:** Instant lookup by patient name or mobile phone number to view their baseline profile and chronic medical history.

- **Prescription & Lab Report Access:** Immediate access to all past digital prescriptions and uploaded diagnostic lab test results.

6. Appointment & OPD Queue

- **Appointment Booking & Tracking:** Slot booking engine aligned with doctor availability, supporting status workflows (Confirmed, Cancelled, Completed).

- **Live Patient Queue:** Token-based queue system for receptionists and nurses to manage patient check-in, waiting room progression, and doctor-room entry.

7. Prescription & Medical Records

- **Digital Prescription Viewer:** Access and view digital prescriptions issued by hospital doctors, including dosage details and medication history.

- **Documents & Discharge Summary:** Cloud storage for diagnostic files, previous reports, and structured patient discharge summaries.

8. Blood Bank Management

- **Real-Time Blood Stock Tracking:** Live inventory tracking by blood group (A+, B+, AB+, O+, and negative counterparts) with component-level categorization (Whole Blood, Packed Red Cells, Platelets, Plasma).
- **Batch & Expiry Monitoring:** Records blood bag collection dates, batch numbers, storage units, and automatic alerts for nearing expiration dates.
- **Emergency Request Broadcast:** One-click module for hospital admins to trigger emergency blood requests to verified donors registered within proximity of the hospital.
- **Transfusion & Issue Log:** Full audit log documenting blood units dispensed, linked directly to the recipient patient's clinical file and attending physician.

8. Hospital Analytics & Reports

- **Patient & Appointment Reports:** Aggregated counts of daily and monthly patient visits and appointment fulfillment rates.

- **Doctor & Department Stats:** Workload reports tracking individual doctor consultation counts and department utilization.

- **PDF/CSV Export:** One-click data export for all administrative and operational reports in standard CSV or PDF format.